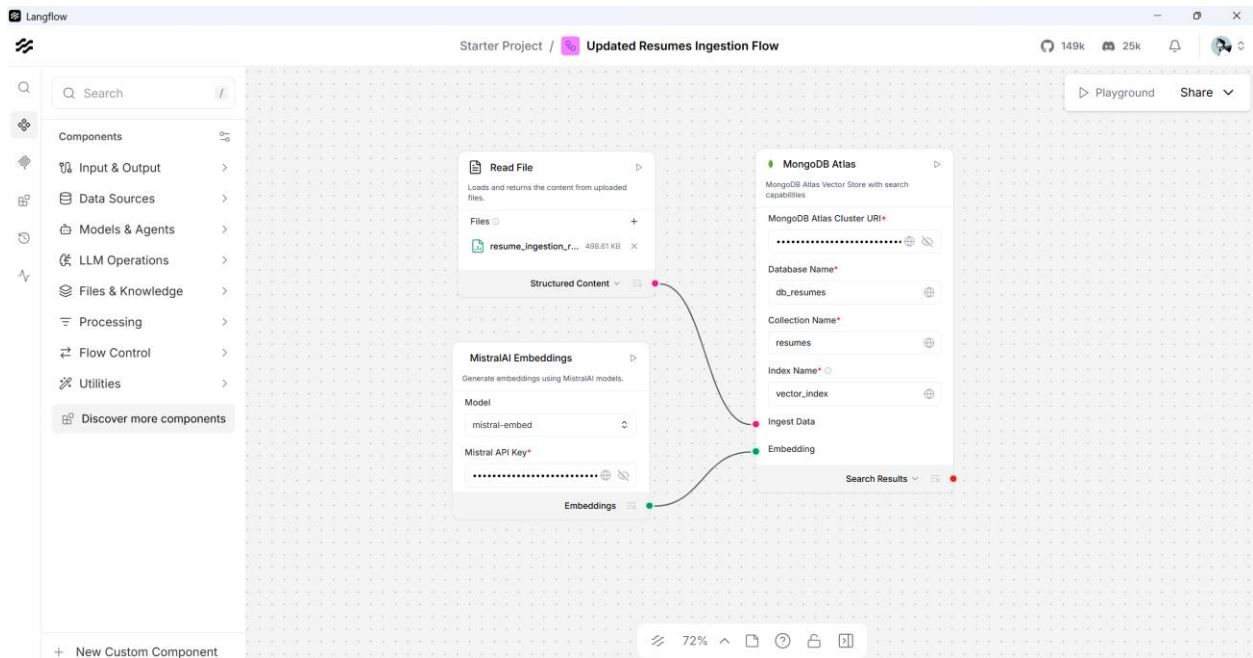


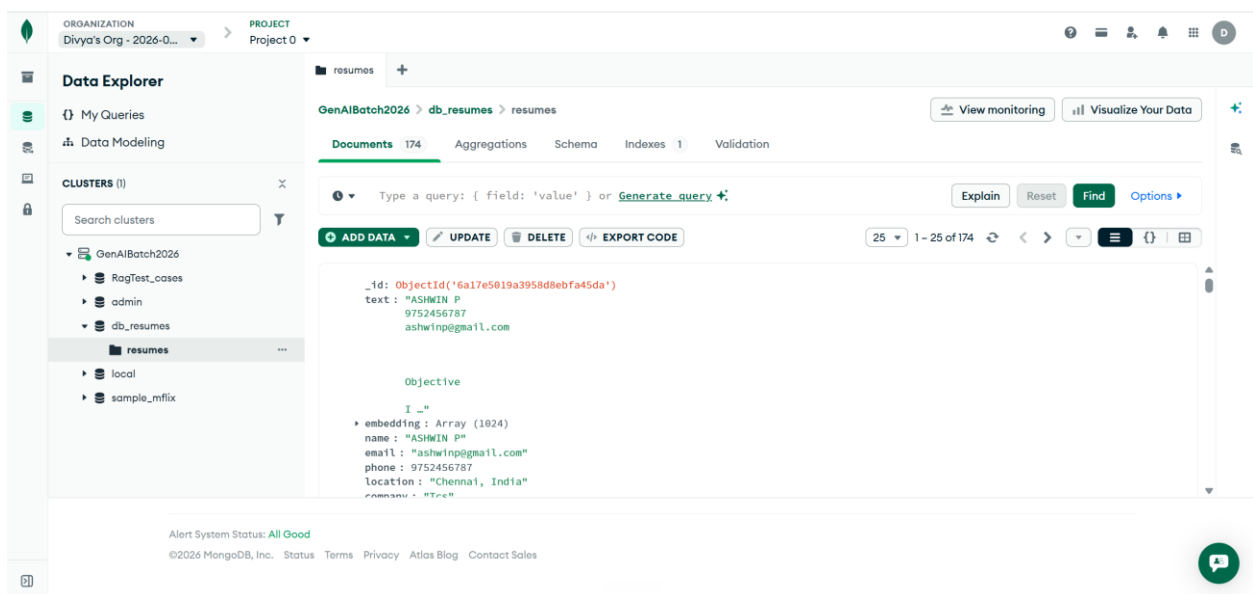
Tools – Week 5 day 2 – Assignment

Complete the Resume RAG, backend implementation and test it

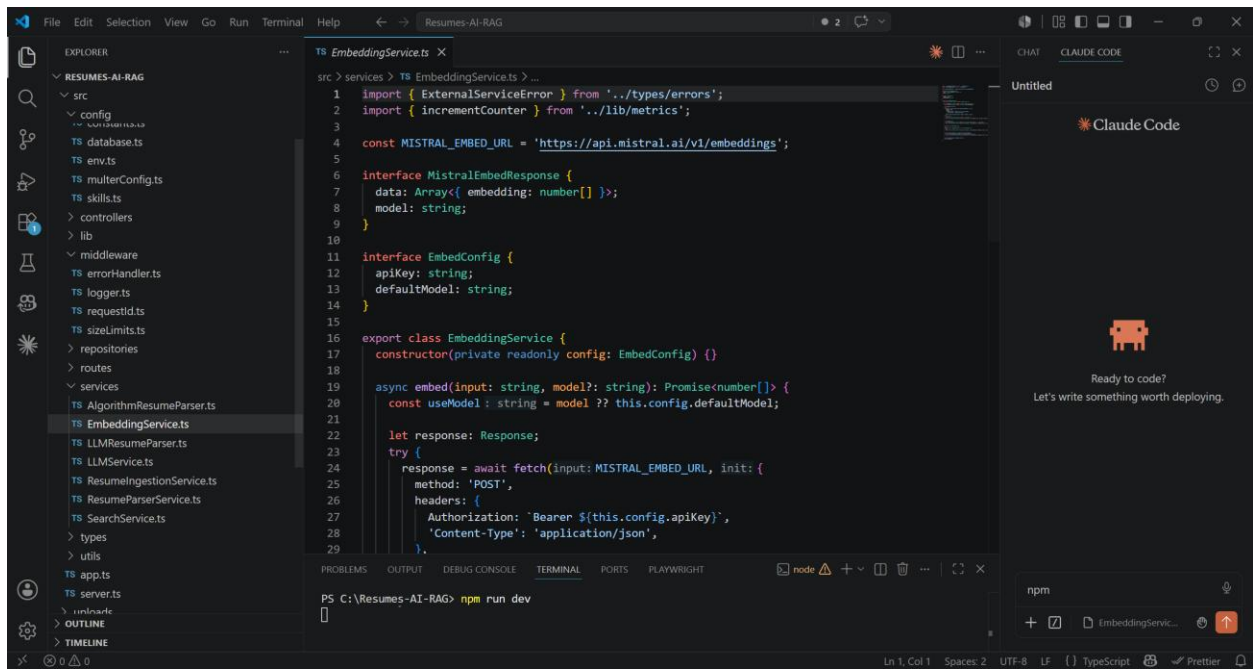
Ingestion flow – Inserted resumes into Mongo DB using the Lang flow (Resume ingestion .json)



Resumes inserted into Mongo DB – resumes folder

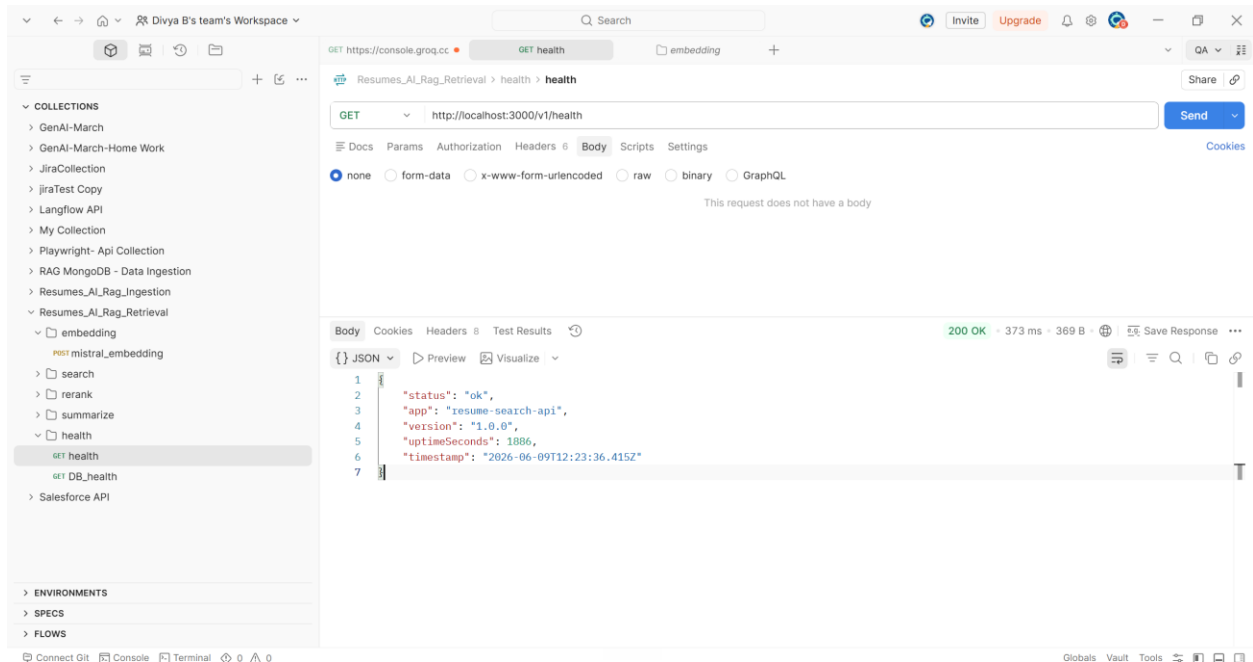


Implemented Resumes RAG – Retrieval and Ingestion in VS code using Claude Code



Imported the collection into Postman and validated the endpoints

Health Check – Success



DB-Health check

The screenshot shows the Postman interface with a workspace named "Divya B's team's Workspace". On the left, the "COLLECTIONS" list includes "Resumes_AI_Rag_Retrieval" > "health" > "DB_health". The main panel shows a GET request to "http://localhost:3000/v1/health/db". The response is a 200 OK status with a latency of 106 ms and a body of 279 B. The JSON response is:

```
{  "status": "ok",  "latencyMs": 83}
```

Embedding

The screenshot shows the Postman interface with a workspace named "Divya B's team's Workspace". On the left, the "COLLECTIONS" list includes "Resumes_AI_Rag_Retrieval" > "embedding" > "mistral_embedding". The main panel shows a POST request to "http://localhost:3000/v1/embeddings" with a JSON body: {"input": "hello world"}. The response is a 200 OK status with a latency of 1.25 s and a body of 19.23 KB. The JSON response is:

```
{  "embedding": [    -0.01953125,    0.0267486572265625,    0.012451171875,    0.028839111328125,    0.04248846875,    0.04327392578125,    0.012647986390625,    0.025299872265625,    -0.022552498234375,    -0.06878143318546875,    -0.02215576171875,    0.0296985187421875,    -0.0285841864453125,    -0.034088134765625,    -0.0443115234375,    0.0046539386640625,
```

BM25 Search

Resumes_AI_Rag_Retrieval > search > BM25_search

POST http://localhost:3000/v1/search/bm25

Body

```
1 {
2   "query": "Selenium Java",
3   "topK": 10
4 }
5
6
```

200 OK - 288 ms - 4.48 KB

Body

```
1 {
2   "results": [
3     {
4       "resumeId": "6a1a9b126ba43ab0cf806c0e",
5       "name": "JAWAHAR RAJENDRAN",
6       "role": "JAVA Developer",
7       "snippet": "JAWAHAR RAJENDRAN\nJava SpringBoot Developer\njawaharraajendran30@gmail.com\n+91-6374894474\nTamil Nadu, Chennai\nwww.linkedin.com/in/jawahar-rajendran-47411b1ab\nLeetCode\nSUMMARY\nJava Developer with 4 years of proven experience in architecting high-performance Multithreaded applications and deploying\nnscale",
8       "score": 1.9745110273361206
9     },
10    {
11      "resumeId": "6a1a9cf7a71d3e2eb73961ca",
12      "name": "JAWAHAR RAJENDRAN",
13      "role": "JAVA Developer",
14      "snippet": "JAWAHAR RAJENDRAN\nJava SpringBoot Developer\njawaharraajendran30@gmail.com\n+91-6374894474\nTamil Nadu, Chennai\nwww.linkedin.com/in/jawahar-rajendran-47411b1ab\nLeetCode\nSUMMARY\nJava Developer with 4 years of proven experience in architecting high-performance Multithreaded applications and deploying\nnscale",
15      "score": 1.9745110273361206
16    }
17  ]
18 }
```

Resumes_AI_Rag_Retrieval > search > BM25_search

POST http://localhost:3000/v1/search/bm25

Body

```
1 {
2   "query": "Selenium Java",
3   "topK": 10
4 }
5
6
```

200 OK - 288 ms - 4.48 KB

Body

```
38 {
39   "results": [
40     {
41       "resumeId": "6a26780b8df29e5f463ebfb9",
42       "name": "P Dineshkumar",
43       "role": "Java Developer",
44       "snippet": "P Dineshkumar\nJava Developer\nEmail - pdineshkumar2407@gmail.com Mobile - +918610849219\nA Java Developer with 6 + Years of IT experience in development of web applications using\nJava Technologies.\nHands on experience programming in J2EE (Servlets, JSP), Core Java and Collections.\nHands on experience w",
45       "score": 1.9515272378921509
46     },
47     {
48       "resumeId": "6a17e5019a3958d8ebfa45f9",
49       "name": "Gayathri Raj",
50       "role": "",
51       "snippet": ". Automation Testing Selenium WebDriver Gayathri Raj Software QA Engineer Proficient in creating and executing test cases, identifying bugs, and enhancing software quality Using tools like Selenium WebDriver, TestNG, Cucumber and JIRA. Hands on experience in Testing web applications. Skilled i",
52       "score": 1.5544523900717163
53     }
54   ]
55 }
```

Vector Search

Resumes_AI_Rag_Retrieval > search > vector_search

POST http://localhost:3000/v1/search/vector

Body

```
1 {
2   "query": "Give me playwright profile",
3   "topK": 10
4 }
5
6
```

200 OK - 804 ms - 4.29 KB

Body

```
1 {
2   "results": [
3     {
4       "resumeId": "6a17e5019a3958d8ebfa4610",
5       "name": "Santhosh G.",
6       "role": "",
7       "snippet": "Santhosh G. Contact Information Address: Dharapuram, Tirupur, India-638656 Phone: +91 9750066477 Email:
8         santhoshg@gmail.com Professional Summary Results-driven QA Engineer with 7+ years of experience in manual and automated
9         testing, specializing in Playwright, Playwright API, performance and functio",
10      "vectorScore": 0.8841888904571533
11    },
12    {
13      "resumeId": "6a17e5019a3958d8ebfa45e0",
14      "name": "Monisha Vijayan",
15      "role": "",
16      "snippet": "Objective: Professional Summary: Experience: Education: Technical Skill Set: MONISHA VIJAYAN Contact: (+91)
17        6381234582 monishvijayan10@gmail.com Accomplished automation test engineer with over 5 years of experience in Functional
18        and Automation Testing. Certified in Playwright with hands -on experi",
19    }
20  ]
21 }
```

Hybrid search

Resumes_AI_Rag_Retrieval > search > hybrid_search

POST http://localhost:3000/v1/search/hybrid

Body

```
1 {
2   "query": "senior node.js backend engineer",
3   "topK": 20
4 }
5
```

200 OK - 575 ms - 16.9 KB

Body

```
1 {
2   "bm25Results": [
3     {
4       "resumeId": "6a266ffdbdf29e5f463ebf9a",
5       "name": "Indhumathi S",
6       "role": "Senior Software Engineer",
7       "snippet": "Indhumathi S\nEmail: indhumailu@gmail.com\nPhone: +91-7092860626 / 9488821051\nLinkedIn: linkedin.com/in/
8         indhumathi-sridharane-775280167\nCareer Objective\nSenior Software Engineer with 6+ years of experience in React and
9         related technologies, aiming into drive innovation and efficiency. Committed to contin",
10      "score": 8.982964515686035
11    },
12    {
13      "resumeId": "6a17e5019a3958d8ebfa45dd",
14      "name": "ANAND G",
15      "role": "",
16      "snippet": "Anand Tirunelveli\n Work as a Data Support Engineer and Automation Test Engineer, an individual contributor
17        and a team player.\n Analyze, Design, Coding and integration in FTPC tool.(Java and SQL).\n Identify and Debug
18        issues with code and suggest changes or improvements.\n Designed, developed.",
19    }
20  ]
21 }
```

Rerank

Resumes_AI_Rag_Retrieval > rerank > llm_rerank_minimal

POST http://localhost:3000/v1/search/rerank

Body

```
1 {
2   "query": "senior node.js backend engineer with MongoDB experience",
3   "candidates": [
4     {
5       "resumeId": "691db80aa895776f97b6eca6",
6       "snippet": "Experienced Node.js backend developer with 6 years building REST APIs and integrating MongoDB...",
7       "score": 2.3
8     },
9   ]
10 }
```

200 OK · 769 ms · 612 B

JSON

```
1 {
2   "results": [
3     {
4       "resumeId": "691db80aa895776f97b6eca7",
5       "snippet": "Fullstack engineer (Node.js, Express) with production MongoDB and replication experience...",
6       "rank": 1
7     },
8     {
9       "resumeId": "691db80aa895776f97b6eca6",
10      "snippet": "Experienced Node.js backend developer with 6 years building REST APIs and integrating MongoDB...",
11      "rank": 2
12    }
13  ],
14  "total": 2,
15  "timings": {
16    "rerankMs": 758
17  }
18 }
```

Rerank Full Content

Resumes_AI_Rag_Retrieval > rerank > **lim_rerank_full_content**

POST http://localhost:3000/v1/search/rerank

Body

```
{
  "id": "691db88aa895776f97b6eca9",
  "name": "ANAND G",
  "role": "Backend Developer API Automation Tester",
  "total_experience": 3,
  "skills": "[\"Java\", \"SQL\", \"Selenium WebDriver\", \"Node.js\", \"Laravel\", \"Postman\", \"MySQL\", \"MongoDB\", \"PostgreSQL\", \"Oracle\", \"Git\", \"Jira\", \"Agile\", \"TestNG\", \"Cucumber\"]",
  "resumeId": "691db88aa895776f97b6eca9",
  "score": 9.348643882751465,
  "snippet": "Anand G is a Backend Developer and API Automation Tester with over 3 years of experience in designing, developing, and testing scalable web applications. He has worked as a Data Support Engineer and Automation Test Engineer, contributing as an individual and team player. His expertise includes analyzing, designing, coding, and integrating in FPMC tool using Java and SQL. He has designed, developed, and maintained automated test scripts using"
}
```

200 OK - 1.68 s - 4.28 KB

Body

```
{
  "results": [
    {
      "resumeId": "691db88aa895776f97b6ecd2",
      "snippet": "Gokul is a Senior Software Engineer with experience in manual and automation QA for mobile applications. He has worked on EcoStruxure Power Device and developed automated scripts for mobile application regression and functional testing using Python with Robot Framework. He has also performed functional testing, UI testing, and regression testing, and is involved in Agile ceremonies. Gokul has experience as a Senior Analyst at Allianz Technologies and as a Test Engineer at Speridian Technologies, where he worked on various testing methodologies and tools.",
      "name": "Gokul",
      "rank": 1
    },
    {
      "resumeId": "691db88aa895776f97b6ecc3",
      "snippet": "Mohan B is a Senior Software Engineer with experience in manual and automation QA for hybrid mobile applications. He has worked on EcoStruxure Power Device at Quest Global, developing automated scripts for regression and functional testing using Python with Robotic Framework and Apium. He has performed functional, UI, and regression testing, and has been involved in Agile ceremonies. Previously, he worked as a Senior Analyst at Allianz Technologies and as a Test Engineer at Speridian Technologies, where he was responsible for functional testing, UI testing, and defect reporting using Jira. He holds a B.E in Automobile Engineering from G.K.M College of Engineering and Technology.",
      "name": "Mohan B",
      "rank": 2
    },
    {
      "resumeId": "691db88aa895776f97b6ecb1",
      "snippet": "VIGNESH S is a Senior QA Engineer with 5.3 years of experience in manual, automation, and API testing. He has worked at Freshworks Private Limited where he designed, developed, and maintained testing artifacts such as Test Plans and Test Strategies using JIRA and Confluence. He performed functional, regression, exploratory, and performance testing on web applications and APIs, and conducted API testing using Postman and Rest Assured. He built and maintained a Selenium WebDriver based automation framework using Java and TestNG, integrated automation test suites into CI/CD pipelines using Jenkins and GitHub, and debugged automation failures. VIGNESH has a B.Tech in Information Technology from Sathyabama University and an MBA in Human Resources from NMIMS.",
      "name": "VIGNESH S",
      "rank": 3
    }
  ]
}
```


Resumes_AI_Rag_Retrieval > rerank > **llm_rerank_full_content**

POST http://localhost:3000/v1/search/rerank

Body

```
{
  "id": "691db80aa895776f97b6eca9",
  "name": "ANAND G",
}
```

200 OK - 1.68 s - 4.28 KB

Body

```
{
  "resumeId": "691db80aa895776f97b6eca9",
  "snippet": "Naveen R is a Test Engineer with experience in both manual and automation testing. He has worked at Srinsoft Technologies since July 2021, focusing on API testing, web automation, and performance testing. He has a background in Mechanical Engineering from Anna University, Chennai. His technical skills include Java, Selenium, Postman, Rest Assured, and JIRA. He has been involved in projects like Enginero and BIMbox Customer Portal, where he designed and implemented automation scripts, performed functional testing, and managed defect reporting.",
  "name": "Naveen R",
  "rank": 5,
}
```

Summarize

Resumes_AI_Rag_Retrieval > summarize > **summarize**

POST http://localhost:3000/v1/search/summarize

Body

```
{
  "query": "Senior Node.js backend engineer with MongoDB experience",
  "candidate": {
    "resumeId": "691db80aa895776f97b6eca6",
    "name": "Ashwin P",
    "snippet": "Built scalable Node.js REST APIs, optimized MongoDB queries, designed sharding and indexing strategies..."
  },
  "style": "detailed",
  "maxTokens": 150
}
```

200 OK - 436 ms - 780 B

Body

```
{
  "resumeId": "691db80aa895776f97b6eca6",
  "name": "Ashwin P",
  "summary": "Here is a 1-2 sentence fit summary for Ashwin P against the job requirement:\n\nAshwin's experience in building scalable Node.js REST APIs and optimizing MongoDB queries makes him a strong fit for this senior Node.js backend engineer role. His expertise in MongoDB sharding and indexing strategies further aligns with the job requirements, demonstrating his ability to handle high-performance and large-scale MongoDB deployments.",
  "timings": {
    "summarizeMs": 425
  }
}
```

End to End

Resumes_AI_Rag_Retrieval > search > end_2_end

POST http://localhost:3000/v1/search

Body

```
1 {
2   "query": "senior node.js backend engineer with MongoDB experience",
3   "topK": 10,
4   "rerankTopK": 8,
5
6   "summarize": true,
7   "summarizeTopK": 3,
8   "summarizeStyle": "short",
9   "summarizeMaxTokens": 150
10 }
```

Body

```
39 {
40   "rank": 5,
41   {
42     "resumeId": "6a266ff98df29e5f463ebf91",
43     "name": "ARUNKUMAR R - Senior Testing Engineer",
44     "role": "Senior Testing Engineer",
45     "snippet": "ARUNKUMAR R - Senior Testing Engineer\nChennai, Tamilnadu | (+91) 9629583892 | raju.arunkumar1985@gmail.com | \nhttps://www.linkedin.com/in/arun-kumar-713b01203/\nPROFESSIONAL SUMMARY\nResults-driven Software QA Engineer with 8 years of experience in manual and automation\ntesting across Cards & Payments(",
46     "rank": 6
47   },
48   {
49     "resumeId": "6a17e5019a3958d8ebfa45dd",
50     "name": "ANAND G",
51     "rank": 7
52   }
53 }
```

Resumes_AI_Rag_Retrieval > search > end_2_end

POST http://localhost:3000/v1/search

Body

```
1 {
2   "query": "senior node.js backend engineer with MongoDB experience",
3   "topK": 10,
4   "rerankTopK": 8,
5
6   "summarize": true,
7   "summarizeTopK": 3,
8   "summarizeStyle": "short",
9   "summarizeMaxTokens": 150
10 }
```

Body

```
18 {
19   "rank": 5,
20   {
21     "resumeId": "6a266ff8df29e5f463ebf9a",
22     "name": "Indhumathi S",
23     "role": "Senior Software Engineer",
24     "snippet": "Indhumathi S\nEmail: indhumailu@gmail.com\nPhone: +91-7092860626 / 9488821051\nLinkedIn: linkedin.com/in/indhumathi-sridarane-775288167\nCareer Objective\nSenior Software Engineer with 6+ years of experience in React and related technologies, aiming\nto drive innovation and efficiency. Committed to contin",
25     "rank": 3,
26     "summary": "Although Indhumathi's background is primarily in React, her 6+ years of experience could translate to a senior backend engineer role, but there's no direct mention of Node.js and MongoDB experience. Further evaluation is needed to confirm her fit for the specific technical requirements of a senior Node.js backend engineer with MongoDB experience."
27   },
28   {
29     "rank": 6,
30     "summary": "Node.js and MongoDB, but her background in backend development and API testing makes her a potential candidate."
31   }
32 }
```

Divya B's team's Workspace

Search

Resumes_AI_Rag_Retrieval > search > end_2_end

POST http://localhost:3000/v1/search

Body

```
1
2 "query": "senior node.js backend engineer with MongoDB experience",
3 "topK": 10,
4 "rerankTopK": 8,
5
6 "summarize": true,
7 "summarizeTopK": 3,
8 "summarizeStyle": "short",
9 "summarizeMaxTokens": 150
10
```

Body

```
62 },
63 "total": 8,
64 "fallbacks": {
65   "bm25Fallback": false,
66   "vectorFallback": false,
67   "rerankFallback": false,
68   "summarizeFallback": false
69 },
70 "timings": {
71   "embeddingMs": 815,
72   "bm25Ms": 134,
73   "vectorMs": 107,
74   "rerankMs": 974,
75   "summarizeMs": 602,
76 }
```

Divya B's team's Workspace

Search

Resumes_AI_Rag_Retrieval > search > end_2_end

POST http://localhost:3000/v1/search

Body

```
1
2 "query": "senior node.js backend engineer with MongoDB experience",
3 "topK": 10,
4 "rerankTopK": 8,
5
6 "summarize": true,
7 "summarizeTopK": 3,
8 "summarizeStyle": "short",
9 "summarizeMaxTokens": 150
10
```

Body

```
64 "fallbacks": {
65   "bm25Fallback": false,
66   "vectorFallback": false,
67   "rerankFallback": false,
68   "summarizeFallback": false
69 },
70 "timings": {
71   "embeddingMs": 815,
72   "bm25Ms": 134,
73   "vectorMs": 107,
74   "rerankMs": 974,
75   "summarizeMs": 602,
76   "totalMs": 2500
77 }
78 }
```